

Integer Data Type and if-else condition

Priority	Operators	Associativity
1 (Highest)	()	Left to Right
2	* / %	Left to Right
3	+ -	Left to Right

<pre>int temp; temp = a; a = b; b = temp;</pre>	<pre>a = a + b; b = a - b; a = a - b;</pre>
<pre>a = a * b; b = a / b; a = a / b;</pre>	<pre>a = a ^ b; b = a ^ b; a = a ^ b;</pre>
<pre>if (condition) { // statements executed when condition is TRUE } else { // statements executed when condition is FALSE }</pre>	<pre>if (condition1) { // statements } else if (condition2) { // statements } else if (condition3) { // statements } else { // statements }</pre>
<pre>if (condition1 && condition2) { // Both conditions are true } else { // One or both conditions are false }</pre>	<pre>if (condition1 condition2) { // One or both conditions are true } else { // Both conditions are false }</pre>
<pre>if ((condition1 && condition2) condition3) { // statements } else { // statements }</pre>	

Attendance %	Marks Awarded
80 – 85%	1
86 – 90%	2
91 – 95%	3
Above 95%	5

```
#include <stdio.h>

void main() {
    int attendance=90;
    int marks;
    if (attendance >= 80 && attendance <= 85) {
        marks = 1;
    } else if (attendance >= 86 && attendance <= 90) {
        marks = 2;
    } else if (attendance >= 91 && attendance <= 95) {
        marks = 3;
    } else if (attendance > 95) {
        marks = 5;
    } else {
        marks = 0;
    }
    printf("Marks Awarded: %d", marks);
}
```

```
#include <stdio.h>

void main() {
    int attendance=97;
    int marks;
    if (attendance > 95) {
        marks = 5;
    } else if (attendance >= 91) {
        marks = 3;
    }
```

```
} else if (attendance >= 86) {  
    marks = 2;  
} else if (attendance >= 80) {  
    marks = 1;  
} else {  
    marks = 0;  
}  
printf("Marks Awarded: %d", marks);  
}
```

Challenge 2

1. Odd or Even
2. Negative or Positive
3. Divisibility Check by 3 and 5

4. Leap Year Checker

- A year is a leap year if it's divisible by 4
- BUT if it's divisible by 100, it's *not* a leap year...
- UNLESS it's also divisible by 400 — then it *is* a leap year

For example, 2024 is a leap year (divisible by 4, not by 100), 1900 is not a leap year (divisible by 100 but not 400), and 2000 is a leap year (divisible by 400).

5. Swap first and last digit of a 3-digit number: 123 → 321
6. Reverse a 3-digit number: 123 → 321
7. Remove the last digit from the given number: 4326 → 432
8. Print the following Grade and Description based on marks

Grade	Marks Range	Grade Point	Description
S	91–100	10	Outstanding
A+	81–90	9	Excellent
A	71–80	8	Very Good
B+	66–70	7	Good
B	61–65	6.5	Above Average
C+	56–60	6	Average
C	50–55	5	Satisfactory
U	< 50	0	Re-appearance

Grade	Marks Range	Description
🏆 S	91–100	Outstanding
😊 A+	81–90	Excellent
👍 A	71–80	Very Good
😊 B+	66–70	Good
😐 B	61–65	Above Average
😐 C+	56–60	Average
😐 C	50–55	Satisfactory
😬 U	< 50	Re-appearance

Sample Output:

If mark=98

🏆 S - Outstanding



Accepted